

Practice Quiz: Planning — Estimating Yardage and Shaping

Part A: Multiple Choice

1. Yardage estimation is best described as: A) Guessing how much yarn to buy B) Prediction under uncertainty using gauge, dimensions, and stitch density C) An exact science D) Something only designers do
2. Planning increases and decreases for shaping is: A) Simple counting B) Model-based computation predicting how stitch changes affect 3D form C) Following instructions without thinking D) Random placement
3. Choosing between top-down and bottom-up construction is: A) Personal preference only B) Selecting between different causal orderings of the generative model C) Irrelevant to the outcome D) Only for garments
4. Buying one extra skein of yarn reflects: A) Wasteful behavior B) Calibrated prediction about estimation uncertainty C) Inability to calculate D) Yarn collection habit
5. Planning for blocking requires: A) No advance thought B) Predicting post-construction fabric transformation from hidden variables C) Following blocking instructions only D) Skipping this step
6. Time estimation for a project requires the model to: A) Only count stitches B) Operate at a higher level of abstraction, predicting the entire project process C) Ignore complexity D) Follow a fixed formula
7. The choice between modular and seamless construction affects: A) Only the appearance B) Error propagation and correction opportunities C) Yarn usage only D) Color choices

Part B: Short Answer

1. You are planning a sweater. Walk through the key planning decisions: yardage estimation, size calculation, construction method, and blocking plan. For each, identify what your generative model must predict and what uncertainties it faces.
2. Explain why a crocheter who weighs their remaining yarn mid-project and recalculates whether they have enough is performing real-time model updating. What sensory data triggers this update?
3. Compare planning a simple scarf (rectangle, one stitch, one size) with planning a fitted garment (multiple pieces, shaping, size options). How does the complexity of the planning problem reflect the complexity of the generative model required?